

Fundamentals of Java

What does a java class contain?

A java class can contain: Variable, Method, Constructor, Static blocks, non-static block

Variables

1. Variable can be defined as a container or placeholder which is mainly used to store data and whatever we store can go and change.
2. Based on type of data variable can further be categorized into two data types
 - Primitive
 - Non-Primitive/Reference

Primitive Data Types

Java supports eight primitive data types

1. Byte
2. Short
3. Int
4. Long
5. Float
6. Double
7. Char
8. Boolean

Observation

1. Whenever we need to print a message of single or multiple words we then go for a pair of double quotes ("").

```
• Class Byte1
{
    public static void main (String [] args)
    {
        byte var=10;
        System.out.println(var);
    }
}
```

2. Whenever we have to print a variable in that doesn't use a pair of double quotes.

2. Variables

Whenever we create any variables, it is required or mandatory that the variable go through two stages.

- Declaration
 - Initialization
1. Comments are used in programming to inform developer or automation tester the reason of writing the code.
 2. Comments are never part of compilation and neither part of execution.
 3. It is not mandatory that in single line we should perform declaration and initialization the variable but it is also possible to declare variable in one and initialize in next line.

```
• class Byte2{
    public static void main(String[] args) {
        byte var1=10; //Declaration and initialization
        System.out.println(var1);
        byte var2;
        var2=20;
        System.out.println(var1);
    }
}

• class Byte3{
    public static void main(String[] args) {
        byte var1;//declaration
        var1=10;//initialization
        var1=20;//Assignment
        var1=30; //Assignment
        System.out.println(var1);
    }
}
```

1. Whatever the value we store in variable that value will get printed
2. Variable at any one of point it store only one value.

```
• class Byte4
{
    public static void main(String[] args)
    {
        byte var1=10, var2=20, var3=30;
        System.out.println(var1);
        System.out.println(var2);
        System.out.println(var3);
    }
}
```

If you have to declare multiple variables of the same time type, we can declare them in single line.

2. Variables

```
• class Byte5
{
    public static void main(String [] args)
    {
        byte var1=10, var2;
        var2=var1;
        System.out.println(var2);
    }
}

• class byte6{
    public static void main(String[] args) {
        byte x=10, y=20,temp;
        System.out.println(x);
        System.out.println(y);
        temp=x;
        x=y;
        y=temp;
        System.out.println(x);
        System.out.println(y);
    }
}
```

IMP Note: Whenever JVM perform execution or when we create variable it only consumes RAM memory.

2. Variables

1. Byte variable in java consumes 1 byte of memory.

1 byte = 8 bits

2. In a computer to measure memory we use a standard called as bits. Bytes and so on.
3. To store values in one byte of memory for byte variable it is calculate below.
- 2^N to $2^N - 1$
- 2^7 to $2^7 - 1$
-128 to 128 -1
-128 to +127

```
• class Byte7
{
    public static void mmain(String [] args)
    {
        byte a=27;
        System.out.println(a);
        byte c=-128;
        System.out.println(c);
    }
}
```

Withing single main method we cannot declare multiple variables of same name.

```
• class Byte8
{
    public static void main(String[] args)
    {
        byte a=100, a=110;
        System.out.println(a);
    }
}
```

2. Variables

Short data type

A short variable consumes 2 bytes memory

Short=2 bytes =16 bits

1

1-16 bits

0-15 bits

-2^n to 2^n-1

-2^{15} to $2^{15}-1$

-32768 to 32767

```
• class short1
{
    public static void main(String[] args)
    {
        short s1=32765;
        System.out.println(s1);
    }
}
```

2. Variables

Int Variable

Integer datatype consumes 4 bytes of memory

Int= 4 byte =4*8

=32 bit

1-32

0-31

-2^n to 2^n-1

-2^{31} to $2^{31}-1$

-2147483648 to 2147483647

```
• class int1
{
    public static void main(String[] args)
    {
        int i1=100;
        System.out.println("i1"+i1);
    }
}
```

1. If have to display a string anything in pair double quotes along with variable then this operation is called concatenation operator.
2. In java the concatenation operation is done help of the + operator.

2. Variables

Long datatype

A long datatype consumes 8 bytes of memory

Long = 8 byte = $8 \times 8 = 64$ bit

1-64

0-63

```
• class long1
{
}
{
    public static void main(String[] args)
    {
        long i1=1000;
        System.out.println("The value of i1: "+i1);
    }
}
```

1. Byte , short , int long they all and store integral numbers (whole number)
Ex:10,15,....
2. Hence byte, short, int, long they commander integral data types.

2. Variables

Identifier

Identifier Naming Rules

1. Name given to a variable, methods, class is called as identifier.
2. A valid identifier in java can start with letter. Currency symbols(\$) and connecting character (-)underscore

```
Int x=293;  
Int $=25;  
Int_=9;  
s.o.p(x);  
s.o.p($);  
s.o.p(_);
```

3. An identifier should never start with number.

```
Int 1=100;  
s.o.p(1);
```

3. We can use numbers after letter, or currency symbol (\$), connecting character underscore.

```
Int x1=100;  
s.o.p();  
int $1=15;  
s.o.p();
```

4. An identifier in java does not have any limit but on global standard length should be between 10-15.
5. Keywords and reserved words cannot be used as valid identifier name.

```
Int byte=8;  
s.o.p();
```

Interview Question:

What are total reserved words or keyword supported by java?

53

6. Class Name or methods names can be used as valid identifier it is supported by java but not considering as good coding practice.

```
Int identifier=10;  
  
Int main=33;  
  
s.o.p(identifier);  
  
s.o.p(main);
```


2. Variables

7. We can declare two identifiers in same area but case of the identifier should be not same only then it is possible to declare identifier with same name.

```
int cyber =10;  
int CYBER=5;  
s.o.p(cyber);  
s.o.p(CYBER);
```

*Valid identifier can contain following

int Cyber25Succes=10;	Valid
int 4Cyber=10;	Invalid
int all@cyber=10;	Invalid
int cyber_success=10;	Valid
int _s =10;	Valid
int cyber#=10;	Invalid
int int=10;	Invalid
int Integer=10;	Valid
int 012_3=10;	Invalid

1. Whenever we assign any to integral datatype the value by default integral.
2. To inform the compiler to treat a value not as integer but long type we need to suffix letter.
3. Java is by default case sensitive but when its come to value java behave like java is case insensitive.

```
• class Example1  
{  
    public static void main(String[] args)  
    {  
        //byte x=128; int to byte  
        //short y=32768; //int to short  
        //int z=12473408934;  
        long a=212434343;  
        System.out.println(a);  
        long b=21474936491;  
        System.out.println(b);  
    }  
}
```

2. Variables

Float and Double

Float and double are made to store decimal value.

FLOAT	DOUBLE
1. If the decimal accuracy is in the range of 5-6 value we go for float. 2. Float can be also defined as single precision. 3. Float consumes 4 bytes of memory.	1. If the decimal 12-14 accuracy, then we go for double. 2. Double can be defined as double precision. 3. Double consume 8 byte of memory.

```
• class Float1
{
    public static void main(String[] args)
    {
        float var1=100.5f; //double to float
        System.out.println(var1);
    }
}
```

Whenever we use float decimal value what is need suffix letter f?

If we don't the suffix letter and by default java will consider decimal value as double type hence ensure the decimal considered as float we have to fix the letter

```
float var2=200.5f;
```

```
S.o.p(var2);
```

Type casting

Typecasting is process of converting one form to another form

It is possible to typecast int to float

It is not possible to typecast float to integer

2. Variables

Char Data Type

1. A character data can be represented as a letter. Number or special character enclosed within pair of ‘ ‘ single quotes

Char:

‘a’

Ex:

Char c1//

2. A character can accept a single letter or number or special character within pair of single quote

//

C++ as well as java both support character datatype however the character in c++ consumes, 1 byte of memory whereas java character consume 2 bytes why?

In C++ technology primarily it support English language

Every English letter number and special character is given in C++ technology is given unique value called as ASCII value (American Standard code of information interchange)

- As java support not only English language but also support other languages but also support other languages hence java consumes more memory with respect to c++ technology.
- Any letter or number as special character internally java gives unique value called as Unicode value
- Unicode is combination of ASCII value + other language unique values

IMP

1. Whenever we assign integer value to character the compiler should generate error but compiler know that integer value and character both are connected or associated with Unicode concept hence it does not throw an error.
2. Whenever we assign int to char or char to int internally the compiler will perform auto typecasting.

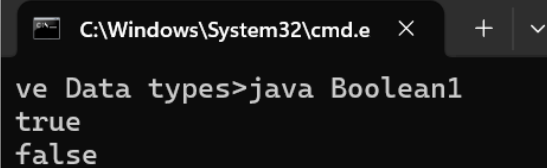
This print statement will display char c but in integer format (Unicode format)

2. Variables

Boolean datatype

1. The Boolean datatype it involves two values True and False.
2. This Boolean datatype will consume one bit of memory but its size is not so accurately defined.

```
1 class Boolean1
2 {
3     public static void main(String []args)
4     {
5         boolean b1=true;
6         System.out.println(b1);
7         boolean b2=false;
8         System.out.println(b2);
9     }
10 }
```

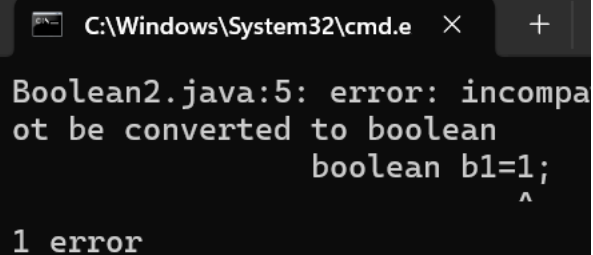


C:\Windows\System32\cmd.e X + v

```
ve Data types>java Boolean1
true
false
```

3. What is output of following code?

```
1 class Boolean2
2 {
3     public static void main(String []args)
4     {
5         boolean b1=1;
6         System.out.println(b1);
7     }
8 }
```



C:\Windows\System32\cmd.e X +

```
Boolean2.java:5: error: incompatible
cannot be converted to boolean
        boolean b1=1;
                   ^
1 error
```

Will Boolean only allow or store data based on particular type hence in java anything apart from true or false will result in error.

2. Variables

Operators

Operators according to precedence or priority.

1. Postfix: `i++`, `i--`
2. Unary: `++i`, `--i`, `+i`, `-i`
3. Multiplicative: `*` / `%`
4. Additive: `+`, `-`
5. Shift: `<<`, `>>`, `>>>`
6. Relational: `<`, `>`, `<=`, `>=`
7. Equality: `==`, `!=`
8. Bitwise AND: `&`
9. Bitwise exclusive OR: `^`
10. Bitwise inclusive OR: `|`
11. Logical AND: `&&`
12. Logical OR: `||`
13. Ternary: `?:`
14. Assignment: `=`, `+=`, `-=`, `<<=`, `!=`

Multiplication and Addition operator

```
• class operator1
{
    public static void main(String[] args)
    {
        System.out.println("Hello World!");
    }
}
```

2. Variables

Interview Questions

1. What does a java class contain?

A java class can contain: Variable, Method, Constructor, Static blocks, non-static block

2. What are the variables?

Variable can be defined as a container or placeholder which is mainly used to store data and whatever we store can go and change.

3. Data types

Primitive data types- byte short int long float double char Boolean

Non primitive data types- String Array Object

4. What are the comments

1. Comments are used in programming to inform developer or automation tester the reason of writing the code.
 2. Comments are never part of compilation and neither part of execution.
-

5. Which memory does variables consume

Whenever JVM perform execution or when we create variable it only consumes RAM memory.

6. Formula to calculate range of variables

-2^N to $2^N - 1$

Range of byte, short

-128 to +127

-32768 to 32767

2. Variables

Data Type	Size	Default Value	Range
<code>byte</code>	1 byte (8 bits)	0	-128 to 127
<code>short</code>	2 bytes (16 bits)	0	-32,768 to 32,767
<code>int</code>	4 bytes (32 bits)	0	-2,147,483,648 to 2,147,483,647
<code>long</code>	8 bytes (64 bits)	0L	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
<code>float</code>	4 bytes (32 bits)	0.0f	~±3.40282347E+38F (6–7 decimal digits)
<code>double</code>	8 bytes (64 bits)	0.0d	~±1.79769313486231570E+308 (15 decimal digits)
<code>char</code>	2 bytes (16 bits)	'\u0000'	0 to 65,535 (Unicode characters)
<code>boolean</code>	JVM-dependent	false	true or false

8. What are the identifiers

Name given to a variable, methods, class is called as identifier.

9. What are the rules to be followed while declaring identifiers

- A valid identifier in java can start with letter. Currency symbols(\$) and connecting character (-)underscore
 - An identifier should never start with number.
 - We can use numbers after letter, or currency symbol (\$), connecting character underscore.
 - An identifier in java does not have any limit but on global standard length should be between 10-15.
 - Keywords and reserved words cannot be used as valid identifier name.
-

9. What are total reserved words or keyword supported by java?

2. Variables

10. Difference between float and double

Float and double are made to store decimal value.

FLOAT	DOUBLE
4. If the decimal accuracy is in the range of 5-6 value we go for float.	4. If the decimal 12-14 accuracy, then we go for double.
5. Float can be also defined as single precision.	5. Double can be defined as double precision.
6. Float consumes 4 bytes of memory.	6. Double consume 8 byte of memory.

11. Whenever we use float decimal value what is need suffix letter f?

If we don't the suffix letter and by default java will consider decimal value as double type hence ensure the decimal considered as float we have to fix the letter

- Whenever we assign any to integral datatype the value by default integral.
 - To inform the compiler to treat a value not as integer but long type we need to suffix letter.
-

12. What is typecasting

Typecasting is process of converting one form to another form

13.C++ as well as java both support character datatype however the character in c++ consumes, 1 byte of memory whereas java character consume 2 bytes why?

In C++ technology primarily it support English language

Every English letter number and special character is given in C++ technology is given unique value called as ASCII value (American Standard code of information exchange)

- As java support not only English language but also support other languages but also support other languages hence java consumes more memory with respect to c++ technology.
 - Any letter or number as special character internally java gives unique value called as Unicode value
 - Unicode is combination of ASCII value + other language unique values
-

2. Variables

